

Coding & Solution

Date	02 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Summary :

Field	Details
Repository Link / URL	https://github.com/John141105/Credit-Card-Approval-Prediction.git
Programming Language(s)	Python 3.11
Framework(s) Used	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Key Features Implemented	• Data Loading and Integration • Data Cleaning and Missing Value Handling • Exploratory Data Analysis (EDA) • Feature Engineering (Age & Employment Years) • Categorical Feature Encoding • Data Scaling using StandardScaler • Train-Test Split • Credit Risk Target Creation • Machine Learning Models (Logistic Regression, Decision Tree, Random Forest, KNN, SVM) • Model Evaluation using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, ROC-AUC • Hyperparameter Tuning using RandomizedSearchCV • Feature Importance Analysis • Model Serialization using Joblib
Pending / Incomplete Features	• Deployment using Flask/Streamlit • Web-based User Interface • Database Integration • Real-time Prediction API • SMOTE-based Class Imbalance Optimization (Optional Enhancement)

Setup / Run Instructions	1. Install Python 3.11 or above.2. Install required libraries using: <pre>pip install pandas numpy matplotlib seaborn scikit-learn joblib</pre> 3. Place <code>application_record.csv</code> and <code>credit_record.csv</code> in the project directory.4. Open the Jupyter Notebook or Python script.5. Execute all cells sequentially.6. The trained model will be saved as <code>credit_card_approval_model.pkl</code> .7. Load the saved model for future predictions.
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Code Quality Checklist :

S. No	Criteria	Status
1	Code is modular and organized into functions/classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments :

The project follows a complete machine learning workflow beginning with data acquisition, preprocessing, and exploratory data analysis, followed by feature engineering, model training, hyperparameter tuning, and performance evaluation. The implementation uses modular and reusable Python functions

to improve maintainability and readability. Multiple machine learning algorithms were trained and compared to identify the best-performing model. Random Forest Classifier achieved the best overall performance for predicting credit card approval status.

The project adheres to good coding practices, including descriptive variable naming, modular function design, code documentation, and reproducible model training. The saved machine learning model can be reused for future predictions and integrated into a web application or REST API for real-world deployment.